FOR A SURJECTIVE OPEN CONTINUOUS MAP THE CLOSURES, CLOSED SETS, OPENS AND LOCALLY CLOSED SETS ARE EXACTLY THE PREIMAGES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The behaviour of closures under a continuous map, and the definition of a submersive map as one whose target carries the quotient topology. Set up in the author's section on submersive maps, immediately before the counted statement. Fixes the four numbered assertions and the notion of submersiveness in the final sentence.

Lemma 0.1. *Let $f : X \to Y$ be surjective, open, continuous map of topological spaces. Let $T \subset Y$ be a subset. Then*

(1) $f^{-1}(\overline{T}) = \overline{f^{-1}(T)}$,
(2) *$T \subset Y$ is closed if and only if $f^{-1}(T)$ is closed,*
(3) *$T \subset Y$ is open if and only if $f^{-1}(T)$ is open, and*
(4) *$T \subset Y$ is locally closed if and only if $f^{-1}(T)$ is locally closed.*

In particular we see that f is submersive.

Proof. It is clear that $\overline{f^{-1}(T)} \subset f^{-1}(\overline{T})$. If $x \in X$, and $x \notin \overline{f^{-1}(T)}$, then there exists an open neighbourhood $x \in U \subset X$ with $U \cap f^{-1}(T) = \emptyset$. Since f is open we see that $f(U)$ is an open neighbourhood of $f(x)$ not meeting T. Hence $x \notin f^{-1}(\overline{T})$. This proves (1). Part (2) is an easy consequence of (1). Part (3) is obvious from the fact that f is open and surjective. For (4), if $f^{-1}(T)$ is locally closed, then $f^{-1}(T) \subset \overline{f^{-1}(T)} = f^{-1}(\overline{T})$ is open, and hence by (3) applied to the map $f^{-1}(\overline{T}) \to \overline{T}$ we see that T is open in $\overline{T}$, i.e., T is locally closed. □

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